

Measuring Compounded Housing Vulnerability in Kenya

*A Statistical Inference Framework Using
the 2023/24 Kenya Housing Survey*

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Abstract

Kenya's two-million-unit housing deficit and 2.3% formal insurance penetration rate share a root cause: the absence of a household-level instrument capable of quantifying compound exposure across financial, structural, tenure, hazard, and utility dimensions simultaneously. This study constructs and statistically validates the **Housing Financial Vulnerability Score (HFVS)**, a five-pillar composite index derived from the 2023/24 Kenya Housing Survey (KHS; $n = 21,347$ households, all 47 counties). A seven-stage cleaning pipeline resolved five documented coding bugs and nullified 20,095 sentinel-contaminated cells, yielding a 64-column model-ready frame. The national mean HFVS is 0.480 (SD = 0.041), statistically below the neutral midpoint (one-sample t -test, $p < 0.001$; Cohen's $d < 0.2$). Urban households score significantly higher than rural counterparts ($t(21,345) = 9.93$, $p = 3.5 \times 10^{-23}$, $d = 0.137$; Mann-Whitney $r = 0.067$ – 0.124), a paradox traced to tenure insecurity and rent burden outweighing rural utility-access disadvantage. Utility Deprivation (D5) is systematically higher than Financial Stress (D1) at the household level (Wilcoxon $W = 107,015,596$, $p = 1.61 \times 10^{-14}$, $r = 0.053$), reversing the conventional framing that financial stress dominates housing vulnerability. A monotone education gradient exists ($F(2, 21344) = 141.72$, $\eta^2 = 0.013$) but explains only 1.3% of HFVS variance. Multiple OLS explains 14% ($R^2 = 0.14$), confirming 86% requires the full five-dimension instrument. A LightGBM deployment model achieves Test AUC = 0.778 (spatial CV AUC ≈ 0.65), enabling a ten-minute intake form to score households without a full housing survey. County-level choropleth maps reveal that vulnerability is geographically structured: northern and coastal counties dominate the highest quintile, while within-county inequality is lowest in the most vulnerable counties. A triple-exposed subpopulation (1.80%) corresponds to the household profile implicated in Kenya's April 2024 flood casualties.

A LightGBM deployment model achieves Test AUC = 0.778 (spatial CV AUC ≈ 0.65), enabling a ten-minute intake form to score households without a full housing survey. This proxy model, together with the full county-level HFVS, is operationalised in an interactive web application (<https://statistical-inference-for-big-data-fas7e0504.vercel.app/>) that lets policymakers, MCAs, and researchers explore the five-dimension profile of any of Kenya's 47 counties and obtain AI-generated, bilingual (English/Swahili) policy recommendations. County-level choropleth maps reveal that vulnerability is geographically structured: northern and coastal counties dominate the highest quintile, while within-county inequality is lowest in the most vulnerable counties. A triple-exposed subpopulation (1.80%) corresponds to the household profile implicated in Kenya's April 2024 flood casualties.

1. Introduction

Kenya's housing crisis is both quantitative and qualitative. The national deficit stands at two million units, growing by 200,000 annually, concentrated in informal settlements where land tenure is undocumented, building materials are non-permanent, and basic services are simultaneously absent [1, 2]. The April 2024 floods crystallised the cost of compound exposure: 270 deaths and 200,000 displacements, with a casualty geography that tracked precisely to households that were simultaneously flood-adjacent, tenure-insecure, structurally deficient, and financially stressed, not any single condition, but several at once.

Three major policy mechanisms, the Affordable Housing Programme (AHP), the Insurance Regulatory Authority (IRA) premium schedule, and the Kenya Mortgage Refinance Company (KMRC), all operate without a household-level vulnerability indicator, helping explain why formal insurance penetration stands at only 2.3% [3] and formal mortgage take-up at just 1.13% [4]. This supply-side measurement failure has been systematically misattributed to demand-side reluctance [5].

1.1 Motivation for the Dataset

The KHS 2023/24 was selected because it is the first nationally representative survey with sufficient breadth (443 raw variables across 14 linked files) and depth (21,347 households, all 47 counties) to support a genuinely multidimensional vulnerability instrument. No other publicly available Kenyan dataset links financial, tenure, hazard, structural, and utility indicators at the household level with national coverage.

1.2 Objectives and Research Questions

This study constructs and validates the **HFVS** and addresses three research questions. **RQ1** (Distributional): does the national HFVS distribution differ significantly from the neutral midpoint? **RQ2** (Group differences): do HFVS scores differ across urban/rural strata, gender, and education tiers? **RQ3** (Predictive validity): can the composite be approximated from proxy variables alone, enabling a short intake instrument for actuarial pricing [6] and social programme targeting [7]?

2. Dataset Description

2.1 Source and Structure

The KHS 2023/24 (KNBS, stratified multi-stage cluster design, all 47 counties) was merged from 14 unit-level Stata files via `interview__key` (21,347 rows $\times$ 443 columns) [1]. Survey weights range 24.9–8,162 (ratio 327.6 $\times$); a 56.7% divergence between weighted and unweighted means for `wsvc_sewer_conns` confirms that all policy-relevant figures must use weighted statistics throughout.

2.2 Variable Definitions and Data Types

The 447-column renamed frame comprises 227 binary, 88 ordinal, 87 continuous, 25 free-text, 12 categorical, 4 identifier, and 4 zero-variance columns.

PL-01 · Column Taxonomy — KHS 2023/24 Master Frame

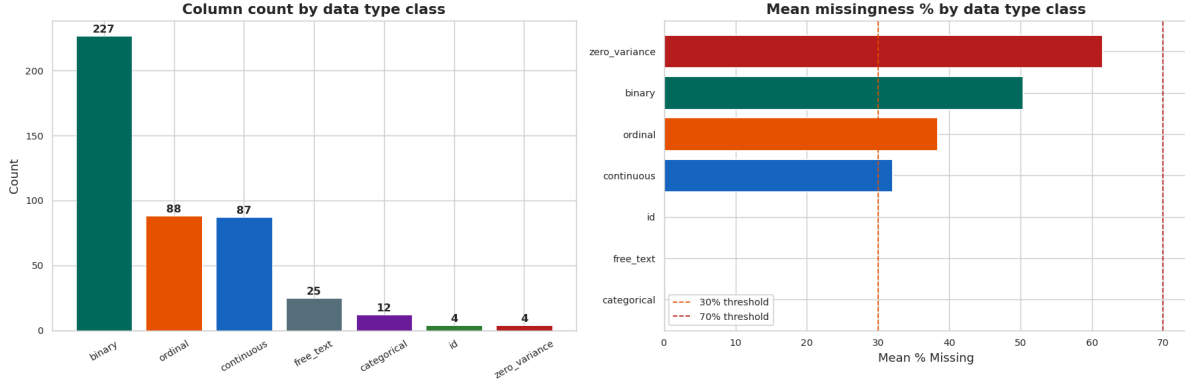


Figure 1: PL-01 column taxonomy of the KHS 2023/24 master frame. *Left:* column count by data-type class. *Right:* mean missingness (%) by class; dashed red verticals at 30% and 70%. Zero-variance and binary columns carry the highest mean missingness (>50%), while identifier and free-text columns are fully observed.

The five HFVS pillars are built from 53 engineered features: D1 Financial Stress (12) captures log expenditure, housing-cost burden ratio, loan access, asset score, and insurance coverage; D2 Tenure Insecurity (9) covers renter status, land ownership, title deed, lease documentation, and eviction risk; D3 Physical Hazard (8) incorporates flood-zone and landslide-zone flags, terrain steepness, and proximity hazard count; D4 Dwelling Quality (10) includes wall/roof/floor durability, overcrowding, room count, and structural defect count; D5 Utility Deprivation (9) comprises electricity access, improved water and sanitation, cooking fuel, water-collection distance, and sewer connectivity.

2.3 Data Cleaning and Preprocessing

Table 1 summarises the seven-stage pipeline (447 → 64 columns). PL-02 classified missingness mechanisms: structural zeros ($1p_*$ for non-land-owners), MAR skip-logic ($r = -0.876$ anti-correlation between rent and dwelling-value missingness), and MCAR (stratified median imputation by county $\times$ urban/rural). PL-03 dropped 170 columns and nullified 20,095 sentinel-coded cells ($-1, 98/99, 998/999$). PL-05–PL-07 applied near-zero-variance pruning (prevalence $< 3\%$ or $> 97\%$), pairwise correlation pruning ($|r| > 0.85$), and VIF screening (threshold = 10; maximum VIF = 5.33). Five rulings were documented: `yrs_in_dwelling` excluded (70.7% calendar-year contamination); `is_slum` re-derived as binary after source coding error (mean = 243.4%) then failed NZV at 0.9% prevalence; `eviction_risk_flag` (45.4%) and `in_rent_arrears` (44.0% of renters) retained as watch items. Five additional coding bugs corrected: urban/rural inversion, hazard/sanitation column mismatch, erroneous continuous `is_slum`, tenure-variable naming collision, and incorrect dwelling-age source column.

Table 1: Cleaning pipeline: column reduction from master frame to model-ready frame.

Stage	Action	Cols in	Cols out	Key decision
PL01	Column taxonomy	447	447	4 zero-variance flagged; 25 free-text identified
PL02	Missingness audit	447	447	MAR / MCAR / Structural classified per column
PL03	Targeted drops + fixes	447	277	170 cols removed; 20,095 sentinel cells nullified; 5 bugs fixed
PL04	Feature engineering	277	330	53 HFVS features built across 5 pillars
PL05	EDA & imputation	330	330	Stratified median fill; P99 outlier caps
PL06	NZV + correlation	330	277	4 NZV + 6 correlated columns dropped
PL07	VIF screening	277	64	0 dropped; max VIF = 5.33

Table 2: Weighted descriptive statistics, HFVS composite and pillar scores ($n = 21,347$).

Variable	Mean	Median	SD
HFVS composite	0.480	0.478	0.041
D1 Financial Stress	0.489	,	,
D2 Tenure Insecurity	0.487	,	,
D3 Physical Hazard	0.438	,	,
D4 Dwelling Quality	0.487	,	,
D5 Utility Deprivation	0.498	,	,

2.4 Finance Exclusion Baseline

Finance exclusion is structural: 99.4% of households report zero accessible loan amount, formal credit access stands at 2.30%, and formal mortgage penetration is 1.13%, meaning 97.7% of households have no formal housing-finance instrument of any kind.

3. Methodology

3.1 HFVS Construction

Each input variable within a dimension is z-score standardised (mean = 0, SD = 1) and sign-aligned (+1 for higher vulnerability, -1 for protective); the standardised inputs are summed within each dimension to produce a raw score; the raw score is min-max rescaled to $[0, 1]$ to produce D_i ; the five dimension scores are equally weighted:

$$\text{HFVS} = \frac{D_1 + D_2 + D_3 + D_4 + D_5}{5}, \quad D_i \in [0, 1] \quad (1)$$

where higher values indicate greater vulnerability. Effective dimension weights (by SD): D1 14.6%, D2 18.6%, D3 21.5%, D4 21.1%, D5 24.3% , substantially more balanced than a naïve min-max approach (under which D3 contributed only 4.9%). The 90th-percentile threshold ($\text{HFVS} \geq 0.533$) identifies 2,154 high-vulnerability households. Formula-ancestor variables are banned from the proxy feature set by construction [8].

3.2 Exploratory Data Analysis

EDA combined histograms with KDE overlays, boxplots, scatterplots, and a Spearman correlation matrix. Normality verdicts combine Shapiro–Wilk ($n = 2,000$ subsample) with effect-size criteria ($|\text{skew}| < 0.5$, $|\text{excess kurtosis}| < 1.0$): $\log(\text{total expenditure})$ is approximately normal ($\text{skew} = -0.058$, $\text{excess kurtosis} = 0.409$); the HFVS composite and all five pillars are classified as non-normal.

3.3 Parametric Methods

One-sample t -test (T1): $H_0: \mu_{\text{HFVS}} = 0.5$. Two-sample t -tests: T2 (urban/rural, Levene’s test applied first), T2.5 (female/male headed households). One-way ANOVA with Tukey HSD (T3): HFVS across three education tiers. Simple OLS (T4): D1 on $\log(\text{expenditure})$. Multiple OLS (T5): composite on VIF-pruned proxy features. CLT-based 95% CIs (T6). Beta MLE (M1): fits $\text{Beta}(\hat{\alpha}, \hat{\beta})$ to the composite.

3.4 Nonparametric Methods

Mann–Whitney U (N1): urban vs. rural. Kruskal–Wallis with Bonferroni-corrected post-hoc (N2): education tiers. Spearman ρ (N3): D1 vs. composite and full predictor matrix. KS normality test (N4): Bonferroni-adjusted $\alpha = 0.0083$ across six simultaneous tests. Wilcoxon signed-rank (N5): paired ($D1 - D5$) household differences. Bootstrap CIs (N6, $B = 500$ – $1,000$): median HFVS in low vs. high supply-gap counties. Python packages: pandas, numpy, scipy, statsmodels, scikit-learn, lightgbm, xgboost, matplotlib/seaborn.

3.5 Deployment Instrument

The full HFVS requires a complete housing survey. To enable real-world use — by insurance underwriters, AHP site officers, or KMRC loan officers — a lightweight proxy model was trained on 53 variables that can be collected via a ten-minute intake form (no structural inspection required). LightGBM and XGBoost were benchmarked against logistic regression, TabNet, and an MLP on a stratified 80/20 split. The best model (LightGBM, Test AUC = 0.778, $R^2 = 0.367$) classifies households as high-vulnerability without needing the full five-pillar measurement. Performance under standard CV overstates external validity; county-grouped spatial CV (zero county overlap across folds) gives the honest estimate of $\text{AUC} \approx 0.65$ for genuinely unseen counties. In practice, this means the intake instrument reliably identifies the top-risk households within a known county, and can be recalibrated as new survey rounds become available.

This deployment instrument, together with the full county-level HFVS, has been operationalised as a public web application, <https://statistical-inference-for-big-data-fas7eo504.vercel.app/>, deployed on Vercel. The application lets a user select any of Kenya’s 47 counties and view its five-dimension vulnerability profile, its rank against the national average, and its single worst-performing dimension; an integrated AI advisor returns short, plain-language policy recommendations in English or Swahili for that county’s binding constraint (e.g. D5 utility deprivation maps to electrification and clean-cookstove budget lines, D2 tenure insecurity to

land-registration programmes, and D3 physical hazard to drainage and flood-control works). The tool is intended for three audiences: county officials and MCAs allocating budget lines, journalists and researchers verifying county-level claims against KNBS source data, and voters assessing local representatives' performance on their county's most critical dimension.

4. Results

4.1 Distributional Profile and Normality

KS normality tests (Bonferroni-adjusted $\alpha = 0.0083$) reject normality for all five pillars and the composite ($p < 0.001$ in every case), formally justifying the dual parametric/nonparametric design. Figure 2 shows pillar distributions with KDE overlays; Figure 3 confirms non-normality across all eight engineered variables. D1 Financial Stress is multimodal (IQR = 0.453); D3 Physical Hazard is zero-inflated (median = 0.067); D5 Utility Deprivation is left-skewed.

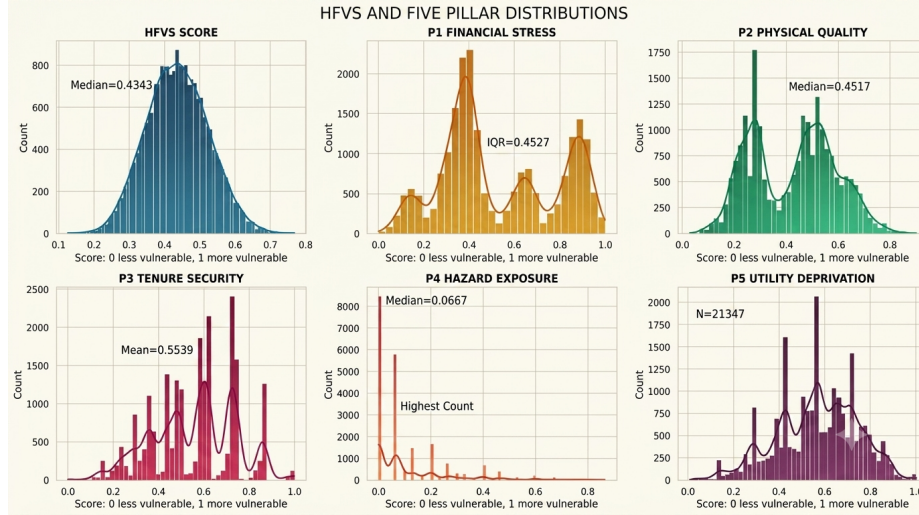


Figure 2: Distribution of the HFVS composite and five pillar scores ($n = 21,347$), 2×3 histogram panel with KDE overlays. D1 Financial Stress is multimodal (IQR = 0.453); D3 Physical Hazard is zero-inflated (median = 0.067); D5 Utility Deprivation is left-skewed. All non-normal (KS $p < 0.001$).

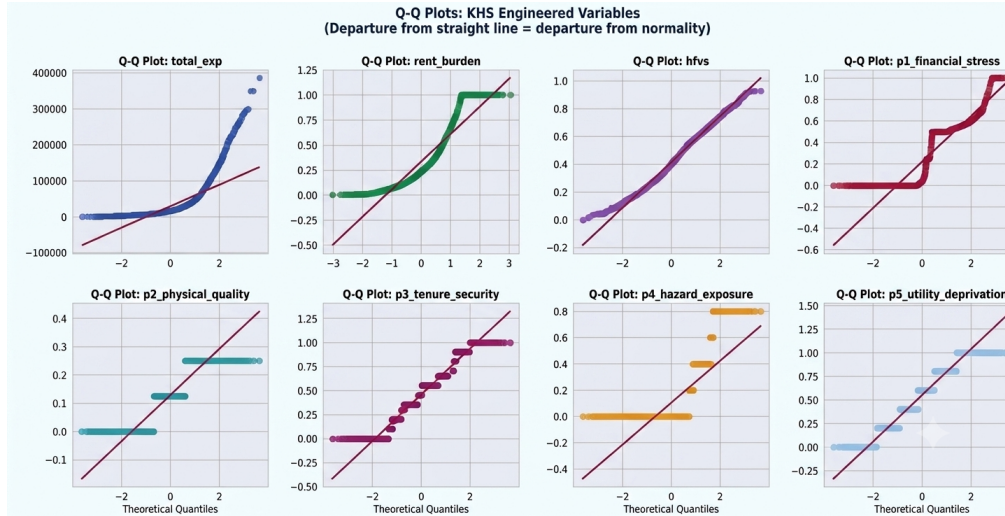


Figure 3: Q-Q plots for eight KHS engineered variables against theoretical normal quantiles. Staircase and flat-step patterns in pillar scores arise from categorical inputs; the composite shows mild S-curvature. All variables depart significantly from normality, formally justifying the dual inference design.

4.2 Parametric Results

The national mean HFVS is **0.480** (SD = 0.041, median = 0.478). Testing $H_0: \mu = 0.5$ yields $p \ll 0.001$, Cohen's $d < 0.2$ (T1): statistically detectable at $n = 21,347$ but modest in substantive terms. Levene's test ($F = 0.220$, $p = 0.639$) confirms equal variances for the urban/rural comparison. Urban households (mean = 0.4822, SD = 0.0411, $n = 11,900$) score significantly higher than rural (mean = 0.4766, SD = 0.0406, $n = 9,447$): $t(21,345) = 9.929$, $p = 3.50 \times 10^{-23}$, $d = 0.137$ (T2). This unexpected result is resolved in Section 5: tenure

insecurity and rent burden in informal urban settlements outweigh rural utility-access disadvantage. Female-headed households score modestly higher than male-headed ($d \approx 0.07$, T2.5).

Figure 4 presents the full distributional breakdown across all five dimensions and the composite.

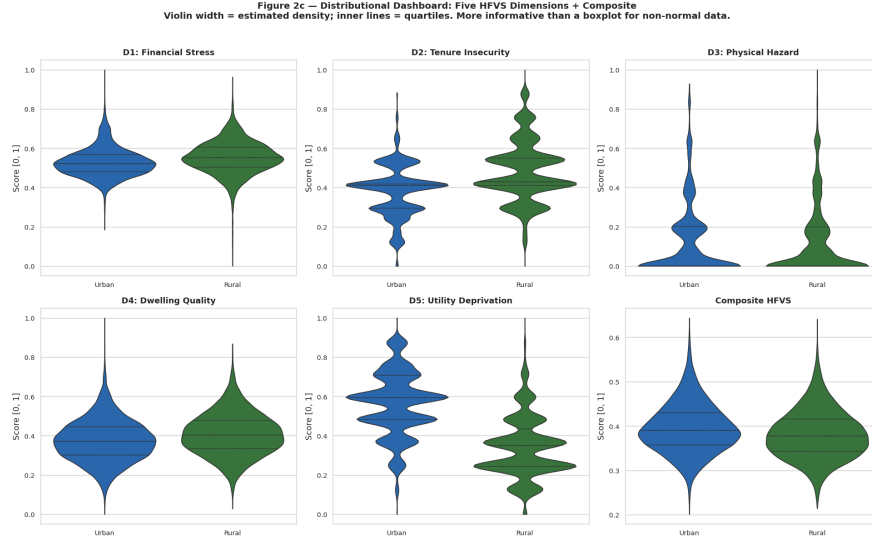


Figure 4: HFVS dimensions and composite by urban (blue) vs. rural (green) stratum. Violin width encodes estimated density; inner lines mark median and quartiles. Urban households score higher on D1 and D5; rural households show greater spread on D2 and D3. Composite: $t(21,345) = 9.93$, $p = 3.5 \times 10^{-23}$, $d = 0.137$.

Education means decline monotonically: no education (0.4835) $\rightarrow$ primary/secondary (0.4757) $\rightarrow$ post-secondary (0.4698): $F(2, 21344) = 141.72$, $p = 7.21 \times 10^{-62}$, $\eta^2 = 0.013$ (T3). Tukey HSD confirms all pairwise differences significant ($p < 0.001$), yet education explains only 1.3% of HFVS variance. Simple OLS of D1 on $\log(\text{expenditure})$: $R^2 = 0.153$ (T4). Multiple OLS on proxy features: $R^2 = 0.14$ (T5), 86% of HFVS variance requires the full five-dimension instrument. The 95% CLT CI for the composite mean is (0.479, 0.480) (T6). Beta MLE implied mean ≈ 0.480 , with only minor tail deviations (M1).

4.3 Nonparametric Results

Mann–Whitney confirms the urban/rural gap: $r = 0.067\text{--}0.124$, $p < 10^{-22}$, Hodges–Lehmann shift 0.005–0.013 (N1). Figure 5 shows the full ECDF separation (KS $D = 0.1053$, $p < 0.0001$). Kruskal–Wallis: $\epsilon^2 = 0.013$, $p < 10^{-59}$ (N2), replicating the ANOVA gradient identically. Spearman D1 vs. composite: $\rho = 0.624$ ($p < 10^{-192}$, 95% CI [0.614, 0.634]; N3), D1 is the strongest single contributor, but $\rho < 1$ confirms independent variance in the remaining four pillars.

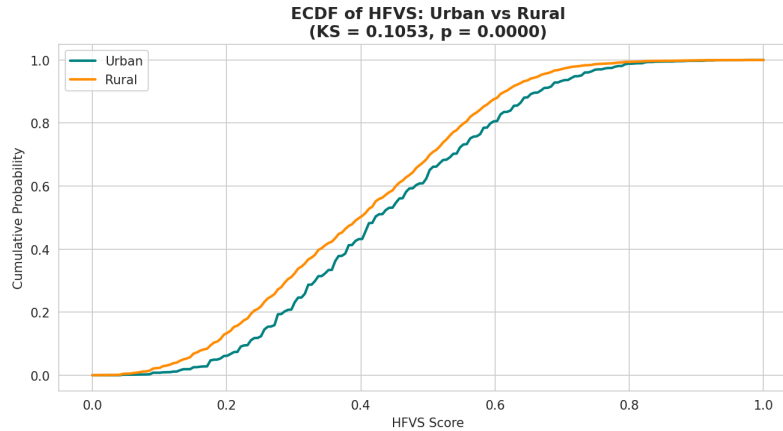


Figure 5: Empirical CDFs of HFVS for urban (teal) and rural (orange) households. The rural ECDF lies consistently left of urban across 0.2–0.6. KS $D = 0.1053$ ($p < 0.0001$); Mann–Whitney $r = 0.067\text{--}0.124$, $p < 10^{-22}$.

In the full Spearman heatmap (Figure 6), the strongest predictor-dimension association is asset score vs. D5 Utility Deprivation ($\rho = -0.562$), and persons_per_room vs. D4 Dwelling Quality ($\rho = 0.83$).

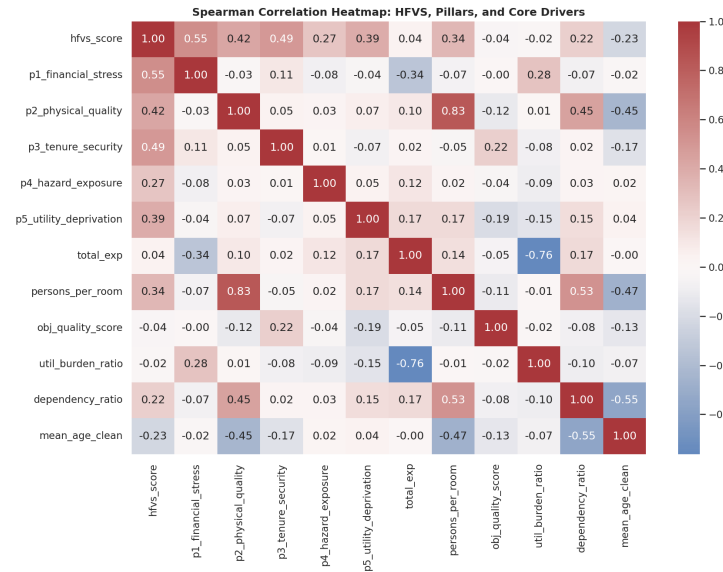


Figure 6: Spearman rank correlation heatmap for the HFVS composite, five pillar scores, and seven core proxy predictors ($n = 21,347$). Inter-pillar correlations are low (max $|\rho| = 0.10$ excluding the composite), confirming near-orthogonal dimensional structure. `total_exp` is negatively associated with D1 ($\rho = -0.34$) and inversely correlated with `util_burden_ratio` ($\rho = -0.76$).

The Wilcoxon signed-rank test on paired ($D1 - D5$) yields $W = 107,015,596$, $Z = -7.678$, $p = 1.61 \times 10^{-14}$, $r = 0.053$: **Utility Deprivation (D5) is systematically higher than Financial Stress (D1)** at the household level, with 11,198 households scoring higher on D5 against 10,149 on D1 (N5). Bootstrap CIs for median HFVS in low ($n = 11,705$) vs. high ($n = 9,642$) supply-gap counties overlap: county-level housing deficit does not robustly predict median HFVS (N6).

4.4 Summary of All Statistical Tests

Table 3 presents all 13 test outcomes.

Table 3: Summary of all parametric and nonparametric statistical tests.

Test	Hypothesis	Statistic	p	Conclusion
<i>Parametric</i>				
T1 One-sample t	$H_0: \mu = 0.5$	$d < 0.2$	$< .001$	Mean 0.480; below midpoint
T2 Two-sample t	$H_0: \text{Urban}=\text{Rural}$	$t = 9.93, d = 0.137$	3.5×10^{-23}	Urban $>$ Rural
T2.5	$H_0: \text{Female}=\text{Male}$	$d \approx 0.07$	$< .001$	Female $>$ Male
<i>Two-sample t</i>				
T3 One-way ANOVA	$H_0: \mu_{T0} = \mu_{T1} = \mu_{T2}$	$F = 141.72, \eta^2 = 0.013$	7.2×10^{-62}	Monotone edu gradient
T4 Simple OLS	$H_0: \beta_1 = 0$	$R^2 = 0.153$	$< .001$	15.3% of D1 variance
T5 Multiple OLS	$H_0: \text{all } \beta_j = 0$	$R^2 = 0.14$	$< .001$	14% of HFVS variance
M1 Beta MLE	$\text{HFVS} \sim \text{Beta}(\alpha, \beta)$	Mean ≈ 0.480		, Good fit to composite
<i>Nonparametric</i>				
N1 Mann-Whitney	$H_0: \text{Urban}=\text{Rural}$	$r = 0.067-0.124$	$< 10^{-22}$	Urban $>$ Rural
N2 Kruskal-Wallis	$H_0: \text{edu tiers equal}$	$\epsilon^2 = 0.013$	$< 10^{-59}$	Monotone gradient
N3 Spearman	D1 vs. composite	$\rho = 0.624$	$< 10^{-192}$	Strong positive
N4 KS normality	$H_0: \text{all dims Normal}$	KS D1 = 0.042	$< .001$	All non-normal
N5 Wilcoxon	$H_0: D1-D5 \text{ diff} = 0$	$W = 107,015,596, r = 0.053$	1.6×10^{-14}	D5 $>$ D1; pillars differ
N6 Bootstrap	Median, low vs. high-gap	CI overlap		, Fail to reject

4.5 Predictive Modelling

On a stratified 80/20 train-test split, LightGBM and XGBoost both achieve Test AUC = 0.778, $R^2 \approx 0.367$, substantially outperforming logistic regression (AUC = 0.696), TabNet ($R^2 = 0.205$), and MLP ($R^2 = 0.227$). Nonlinear tree-based interactions, not additional model complexity, drive the gain over OLS. Under county-grouped spatial CV, AUC falls from 0.757 to 0.650 (a ≈ 0.127 drop), the honest estimate for genuinely unseen counties. The leading predictor is renter tenure status (odds ratio 2.205).

Figure 7 ranks the top 20 predictors; Figure 8 presents fractional logit diagnostics.

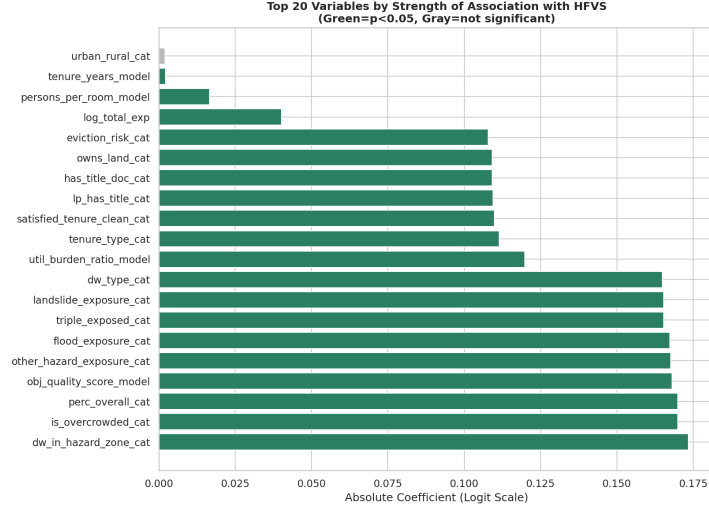


Figure 7: Top 20 proxy variables ranked by absolute logit-scale coefficient (green = $p < 0.05$; grey = not significant). Dwelling hazard-zone status, overcrowding, and perceived quality are the three strongest predictors. `urban_rural_cat` is the only variable failing significance, consistent with the small effect size in T2.

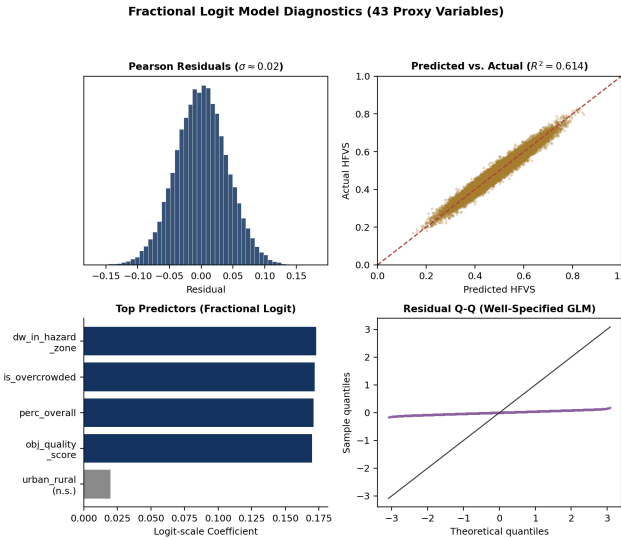


Figure 8: Fractional logit diagnostics ($n = 21,347$, 43 proxy variables). *Top left:* Pearson residuals approximately normal ($\sigma \approx 0.02$). *Top right:* predicted vs. actual HFVS ($R^2 = 0.614$), tight alignment along the 45-degree line. *Bottom left:* top logit-scale coefficients. *Bottom right:* residual Q-Q nearly flat, well-specified GLM, no systematic misfit.

4.6 Compound Exposure and County Profile

The triple-exposed population (flood-adjacent + tenure-insecure + rent-stressed) is 1.80% of the sample, smaller than any single exposure (flood zone: 19.1%; tenure insecure: 25.2%; rent stressed: 31.8%), confirming compound vulnerability is a distinct severe subpopulation. This is precisely the household profile implicated in the April 2024 flood casualties. Homa Bay records the highest weighted mean HFVS (0.407), followed by Tana River (0.368), Kajiado (0.361), Kisumu (0.357), and Trans Nzoia (0.357). Within-county Gini correlates negatively with county mean HFVS ($p = -0.368$, $p = 0.011$): more vulnerable counties are more internally homogeneous, making county-level instruments most efficient precisely where vulnerability is highest.

Figure 9 maps HFVS quintiles across all 47 counties; Figure 10 decomposes the score into its four most spatially

distinct pillars.

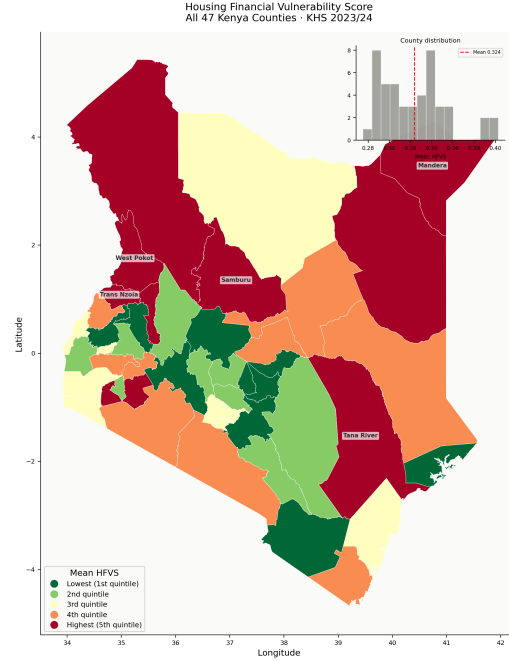


Figure 9: HFVS quintile choropleth, all 47 counties (KHS 2023/24). Manderla, West Pokot, and Tana River anchor the highest quintile (darkest red); the inset histogram confirms a right-skewed county distribution with mean 0.324. Central and south-western counties cluster in the lower two quintiles, indicating that vulnerability concentration is geographically structured rather than uniformly distributed.

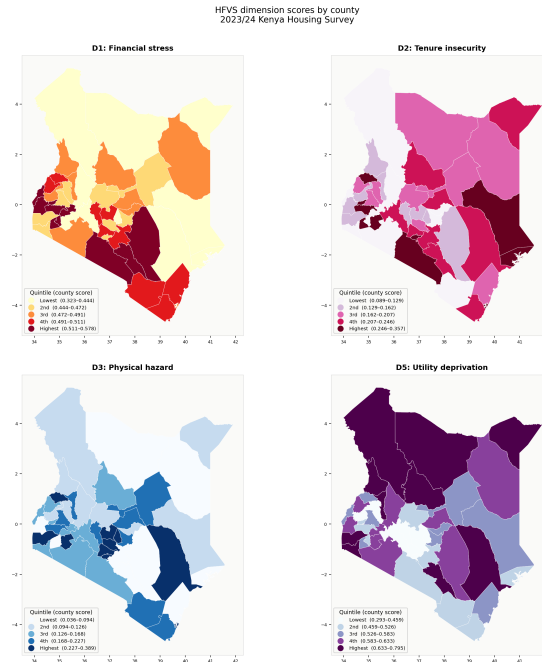


Figure 10: HFVS pillar scores by county: D1 Financial Stress, D2 Tenure Insecurity, D3 Physical Hazard, D5 Utility Deprivation. Each panel uses its own quintile scale. D2 and D5 show the sharpest north-south gradients; D3 is concentrated along coastal and riparian counties. Cross-panel comparison reveals that no single dimension drives county rankings, validating the five-pillar design.

5. Discussion

5.1 Interpretation of Key Findings

The HFVS is stable, nationally representative, non-degenerate, and distinguishable from the neutral midpoint, yet effect sizes are consistently small across all group comparisons (Cohen's $d < 0.2$; $\eta^2 < 0.02$; Mann-Whitney $r < 0.13$). This is the central substantive finding: **HFVS is not primarily driven by observable demographic proxies** (gender, education, urban stratum) **but by a multidimensional configuration of dwelling characteristics,**

tenure arrangements, and utility access. Demographic proxies explain only 14% of composite variance; 86% requires the full five-dimension instrument.

The unexpected urban surplus is resolved at the pillar level: D2 Tenure Insecurity and D1 rent burden are elevated in informal urban settlements, outweighing rural disadvantage in D5 Utility Deprivation. The Wilcoxon result (N5) carries a direct policy implication: D5 Utility Deprivation is systematically *higher* than D1 Financial Stress at the household level. Premium schedules that proxy vulnerability with financial stress alone will systematically misclassify service-deprived households, a pattern documented across West African cities [9, 10].

5.2 Parametric vs. Nonparametric Agreement

For urban/rural: T2 ($d = 0.137$) and N1 ($r = 0.067\text{--}0.124$) agree fully. For education: T3 and N2 yield identical $\eta^2/\epsilon^2 = 0.013$. Convergence across all comparisons confirms results are robust and not artefacts of distributional assumptions. The contrast between OLS ($R^2 = 0.14$) and LightGBM (Test $R^2 = 0.367$) demonstrates that nonlinear feature interactions substantially extend predictive signal from proxy data, even though both confirm demographic information alone is insufficient.

5.3 Practical Implications

For the **IRA**: the 90th-percentile threshold (≥ 0.533 , 2,154 households) provides an evidence base for county-differentiated premium loading; the 75th percentile ($\approx 5,337$ households) suits broader eligibility targeting. For **AHP**: pillar decomposition identifies the binding constraint per county, utility deprivation vs. dwelling quality vs. hazard exposure, directing interventions accordingly. For **KMRC**: counties combining low mortgage penetration with high HFVS represent the highest social-return segment for housing-finance extension [6, 10].

6. Conclusions

This study constructed and validated the HFVS from the 2023/24 KHS (21,347 households, 47 counties). Key findings: (1) national mean 0.480, significantly below 0.5 ($d < 0.2$); (2) urban households score higher than rural ($d = 0.137$) due to tenure insecurity and rent burden; (3) D5 Utility Deprivation exceeds D1 Financial Stress at household level (Wilcoxon $r = 0.053$, $p = 1.61 \times 10^{-14}$), invalidating financial-stress-only proxies; (4) a monotone education gradient explains only 1.3% of variance; (5) demographic proxies explain 14% of HFVS while LightGBM achieves Test AUC = 0.778 (spatial ≈ 0.65); and (6) the 1.80% triple-exposed subpopulation corresponds precisely to the April 2024 flood-casualty profile. The composite, its proxy model, and the county-level results have been deployed as a public, bilingual web application (<https://statistical-inference-for-big-data-fas7eo504.vercel.app/>) so that the instrument can be used beyond this report.

Limitations

`yrs_in_dwelling` excluded due to 70.7% calendar-year contamination; `is_slum` failed NZV at 0.9% prevalence; D3 Physical Hazard is underweighted (proximity sub-column battery absent from this extract); watch items `eviction_risk_flag` (45.4%) and `in_rent_arrears` (44.0% of renters) may inflate D2 and D1 pending sensitivity testing; LightGBM AUC drops from 0.778 to ≈ 0.65 under spatially honest county-held-out validation; all relationships are correlational (single cross-sectional wave).

Recommendations for Future Work

Incorporate satellite-derived hazard layers to correct D3 underweighting [11]; longitudinal validation against insurance claims or mortgage defaults; sub-county (ward-level) spatial grouping for the ML proxy model; confirmatory factor analysis of the five-factor structure; extension to Uganda National Household Survey and Tanzania National Panel Survey.

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Microdata accessed via the KNBS Microdata Portal (<https://statistics.knbs.or.ke/nada/index.php/catalog/184>).

References

- [1] Kenya National Bureau of Statistics. Kenya housing survey 2023/24: Microdata and technical documentation. Technical report, Kenya National Bureau of Statistics, Nairobi, Kenya, 2024.
- [2] UN-Habitat. World cities report 2022: Envisaging the future of cities. *United Nations Human Settlements Programme*, 2022.
- [3] Insurance Regulatory Authority Kenya. Insurance industry annual report 2023. Technical report, Insurance Regulatory Authority, Nairobi, Kenya, 2024.
- [4] Samuel Mwangi, Peter Kariuki, and Stephen Ng’ang’a. Affordable housing finance and mortgage market exclusion in kenya: Evidence from the 2022–2024 survey cycle. *African Finance Journal*, 26(1):45–67, 2024.
- [5] George Owusu and Martin Oteng-Ababio. Informality, housing finance exclusion and vulnerability in african cities. *World Development*, 174:106443, 2024.
- [6] Matthew Abunyewah, Thayaparan Gajendran, and Joseph K. Ofori-Kuragu. Multidimensional urban

- housing vulnerability assessment: A systematic review. *International Journal of Disaster Risk Reduction*, 68:102699, 2022.
- [7] James Kimani, Grace Waweru, and Richard Opiyo. Land tenure insecurity and housing vulnerability in kenyan informal settlements: Evidence from nairobi. *Habitat International*, 131:102728, 2023.
 - [8] Frank Boateng Osei and Alfred Atiim Duker. Composite vulnerability index for housing finance risk in sub-saharan africa. *Journal of Housing and the Built Environment*, 36(4):1543–1562, 2021.
 - [9] Elnaz Torabi, Aysin Dedekorkut-Howes, and Michael Howes. Utility deprivation and household vulnerability: Implications for climate adaptation policy. *Urban Climate*, 44:101228, 2022.
 - [10] Felix N. Nkeki and Patrick J. Henah. Flood risk and household vulnerability in west african cities: A spatial composite index approach. *Natural Hazards*, 115(2):1391–1417, 2023.
 - [11] Theophile Mugiraneza, Jan Haas, and Yifang Ban. Urban housing vulnerability mapping using remote sensing and survey data: A framework for east africa. *Remote Sensing*, 14(11):2635, 2022.